Document Title		Power Monitoring System UMPSA			
		Ilot Substation			
Doc Type	Word	Document type	1	Date	19 December 2025

اونيۏرسيتي مليسيا ڤهڠ السلطان عبد الله
UNIVERSITI MALAYSIA PAHANG
AL-SULTAN ABDULLAH

FACULTY OF TECHNOLOGY ENGINEERING ELECTRIC AND ELECTRONIC

BVI 2232 Application System Development I

Report Project

PROJECT TITLE: Power Monitoring System University Malaysia Pahang Al-Sultan Abdullah

SUPERVISOR: Ts. Dr. Md Rizal Bin Othman

No	Name	ID Matric
1	Ahmad Syahidul Amin Bin Mohd Aznal	VC23054
2	Syamsul Arifin Bin Awi Bowo	VC23047
3	Muhammad Amirul Adly Bin Mohamad Rashidi	VC23053

SUBMISSION DATE: 13 January 2026

TABLE OF CONTENTS

List of Figure

Document Title		**Power Monitoring System UMPSA**			
		IIot Substation			
Doc Type	**Word**	**Document type**	**1**	**Date**	**19 December 2025**

1. Project Introduction

1.1 Project Overview

Electrical energy is one of the most critical operational resources in higher education institutions, particularly for large campuses such as University Malaysia Pahang Al-Sultan Abdullah (UMPSA). The university relies heavily on electricity to support academic buildings, laboratories, administrative offices, hostels, data centers, and various campus facilities. With increasing campus development and growing electrical loads, effective monitoring and management of electrical energy consumption have become essential to ensure operational efficiency, financial sustainability, and environmental responsibility.

Traditionally, electricity billing and monitoring at large facilities are performed through monthly utility bills provided by Tenaga Nasional Berhad (TNB). However, this approach does not provide real-time visibility of energy usage, demand patterns, or early cost estimation. As a result, management decisions related to energy efficiency and budgeting are often reactive rather than proactive.

The advancement of Industrial Internet of Things (IIoT) technology has enabled real-time data acquisition, communication, and analytics for industrial and utility applications. By integrating smart meters, programmable logic controllers (PLCs), human-machine interfaces (HMIs), communication protocols, and cloud-based analytics, a comprehensive energy monitoring system can be developed to provide continuous insights into electrical consumption and cost trends.

This project focuses on the development of an IIoT-based Power Monitoring System for UMPSA, designed to monitor voltage, current, power demand, and energy usage across the entire campus. The system also calculates monthly electricity costs based on actual TNB tariff structures and provides forecasting outputs to support management decision-making.

1.2 Problem Statement

UMPSA currently faces several challenges related to campus-wide electrical energy monitoring and cost management:

1. The absence of a real-time energy monitoring system limits visibility into instantaneous electrical parameters such as voltage, current, and power demand.
2. Monthly electricity cost estimation relies heavily on historical billing data, which does not reflect current consumption trends.
3. High electricity demand charges and network charges contribute significantly to total electricity costs, yet demand patterns are not continuously monitored.
4. The lack of centralized data storage and visualization makes it difficult to analyze energy usage by region or incomer.
5. Management has limited tools to forecast monthly electricity expenditure in advance, affecting budget planning and energy optimization initiatives.

These limitations highlight the need for a centralized, intelligent, and real-time power monitoring solution that integrates industrial devices with IIoT technologies.

1.3 Project Objectives

The main objectives of this project are:

- To design and develop an IIoT-based power monitoring system for UMPSA.
- To monitor real-time electrical parameters including voltage, current, power, and energy consumption.
- To collect energy data from Digital Power Meters (DPMs) using Modbus TCP and Modbus RTU protocols.
- To integrate PLC, HMI, and IIoT communication for centralized monitoring.
- To calculate electricity costs based on TNB tariff structures including peak, off-peak, demand, and network charges.
- To forecast monthly electricity costs for management planning.
- To visualize energy consumption and cost data through a web-based dashboard.

1.4 Project Scope

The scope of this project includes:

- Monitoring electrical energy consumption for the entire UMPSA campus.
- Integration of DPMs installed at TNB panels and sub-distribution boards.
- Use of WECON LX6C-0808MT PLC for data processing.
- Visualization via WECON PI3070IG HMI.
- Data transmission using the MQTT protocol.
- Data storage using MySQL database managed through phpMyAdmin.
- Use Virtual Machines of Tenaga VM Server for the backend and frontend of the system
- Dashboard development for real-time and historical analysis.

This project does not include physical modification of electrical distribution systems or billing reconciliation with official TNB invoices.

1.5 Limitation of the Project

Despite its successful implementation, this project has several limitations. The forecasting method is based on historical consumption trends and does not account for sudden operational changes such as special events, equipment failure, or extreme weather conditions. Additionally, the system relies on continuous network connectivity for IIoT data transmission, which may affect real time monitoring during network outages.

Furthermore, the cost calculation is dependent on predefined TNB Tariff parameters. Any changes to tariff structure would require manual update to the system configuration

1.6 Significance of the Project

The Purpose system provides significant benefits to UMPSA by enabling Real Time Energy Monitoring, Early Cost Estimation, and Data Driven decision making. The System Support Energy Efficiency initiatives, improve operational transparency, and contributes to sustainable campus management aligned with smart campus objective.

2. System Overview

The Power Monitoring System is designed as an end-to-end solution that integrates field measurement devices, control systems, communication networks, backend servers, and a web-based visualization platform. The system begins at the electrical distribution level, where energy parameters are measured by digital power meters installed at various incomers and regions within the campus.

These measurements are collected and consolidated through a SCADA panel in PLC, which acts as an intermediary between the field devices and higher-level control systems. The processed data is then transmitted to a Wecon LX6C PLC, which further manages data distribution to the HMI. The HMI not only provides local visualization but also serves as a gateway to transmit data wirelessly to the central server using MQTT communication.

On the server side, all applications are hosted within a virtual machine, where Node-RED handles data reception and processing, MySQL manages data storage, a PHP-based API provides data access, and an HTML dashboard presents the information to users. This layered architecture ensures scalability, reliability, and ease of maintenance.

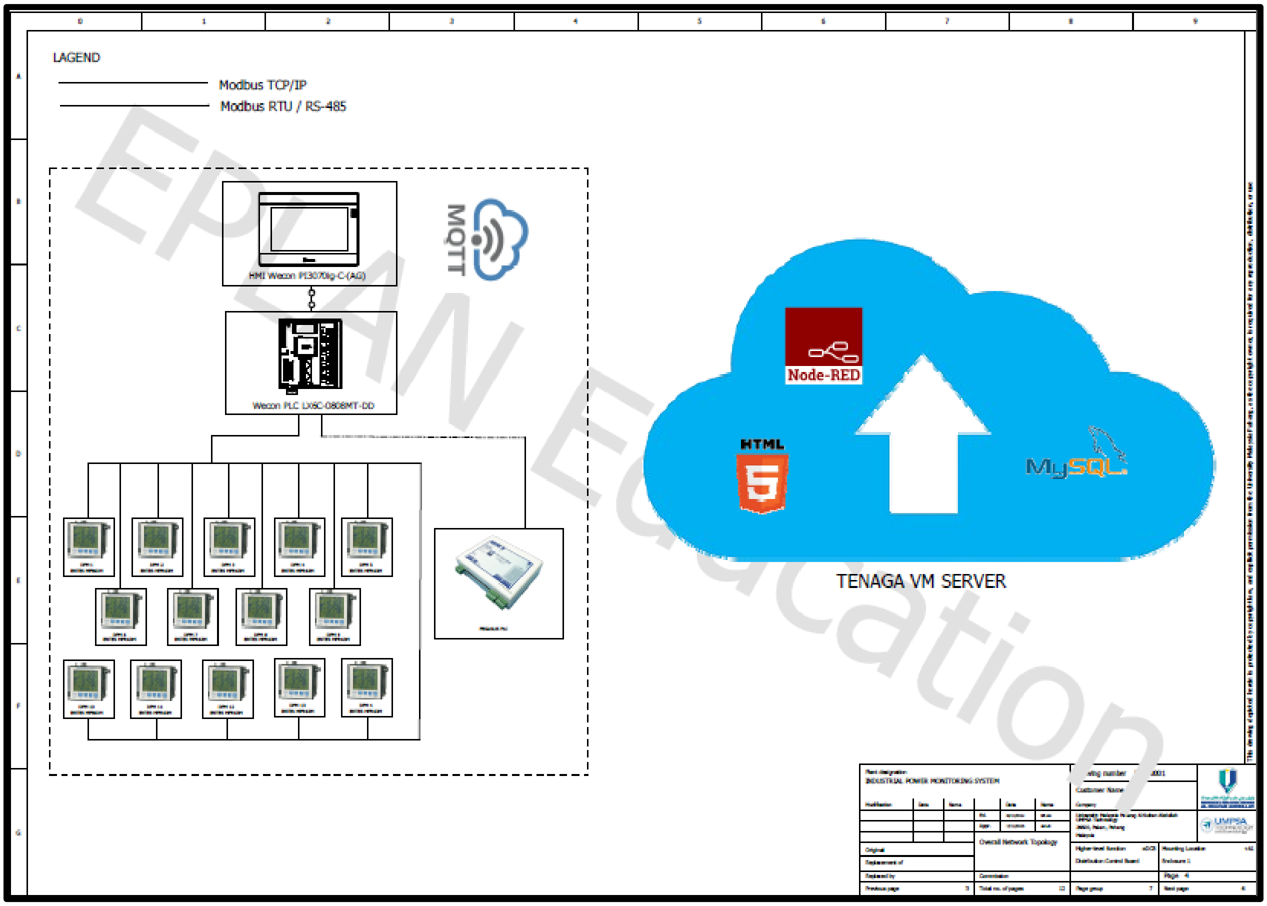


Figure 1 Drawing Topology Project

Figure 1 illustrates the overall architecture of the IIoT-based Power Monitoring System. Electrical parameters are measured using multiple Digital Power Meters (DPM) installed at various incomers and regions. The meters communicate with the WECON PLC via Modbus RTU (RS-485) for centralized data acquisition. The collected data is displayed locally on the WECON HMI and transmitted to the cloud using MQTT. On the server side, Node-RED processes the data and stores it in a MySQL database, while a web-based HTML dashboard enables real-time monitoring and analysis through the Tenaga VM Server.

2.1 System Overview

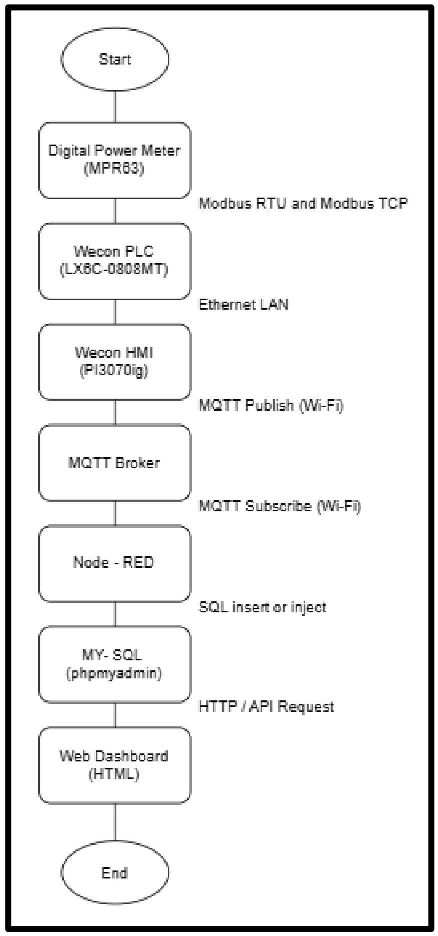


Figure 2 Full System Overview

Figure 2 The provided image provided a whole sequence from pick up the raw value until it shown in web dashboard using all the communication layers, field devices and data storage.

3. Communication Architecture

3.1 Modbus RTU Communication

In the proposed Power Monitoring System, Modbus RTU is used as the primary communication protocol at the field level between the digital power meters (DPMs) and the SCADA panel PLC. A total of seven DPMs are installed at various incomers and regions within the UMPSA Campus Pekan electrical distribution network. These meters continuously measure electrical parameters such as active energy, reactive energy, voltage, current, power factor, and maximum demand.

All DPMs are interconnected using RS-485 two-wire communication in a multi-drop topology, which allows multiple meters to share a single communication line. Each DPM is assigned a unique Modbus slave ID. The SCADA panel PLC acts as the Modbus RTU master, sequentially polling each DPM by requesting specific register addresses. Through this polling mechanism, the SCADA PLC retrieves data from all connected meters in a controlled and determined manner. An RTU converter is used to support stable communication over the two-wire network while maintaining signal integrity across multiple devices.

After collecting data from all DPMs, the SCADA panel PLC processes and maps the required values into its internal memory registers. In this system, energy-related parameters, such as total active energy and cumulative energy consumption, are made available to the Wecon LX6C PLC via Modbus RTU. The Wecon PLC reads these energy registers directly from the SCADA PLC using RTU communication. This design ensures that billing-critical and cumulative energy data is obtained through a robust and noise-immune RS-485 communication path, while preserving a simple wiring structure and minimizing installation complexity and cost.

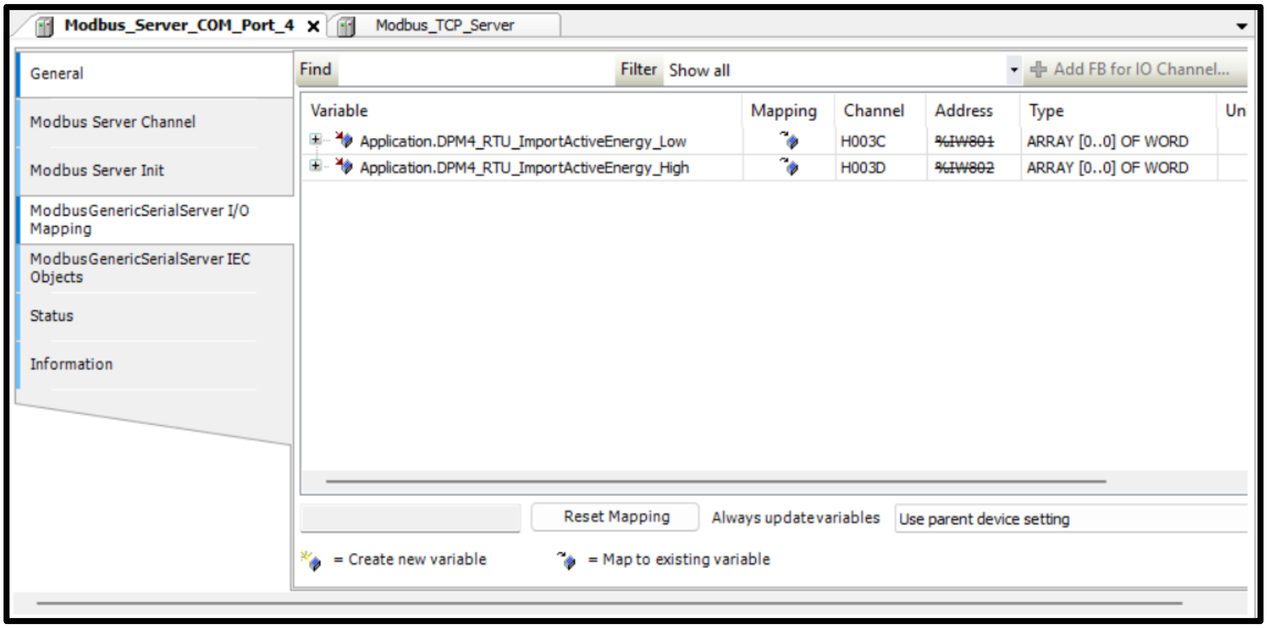


Figure 3 Parameters used Modbus RTU Server

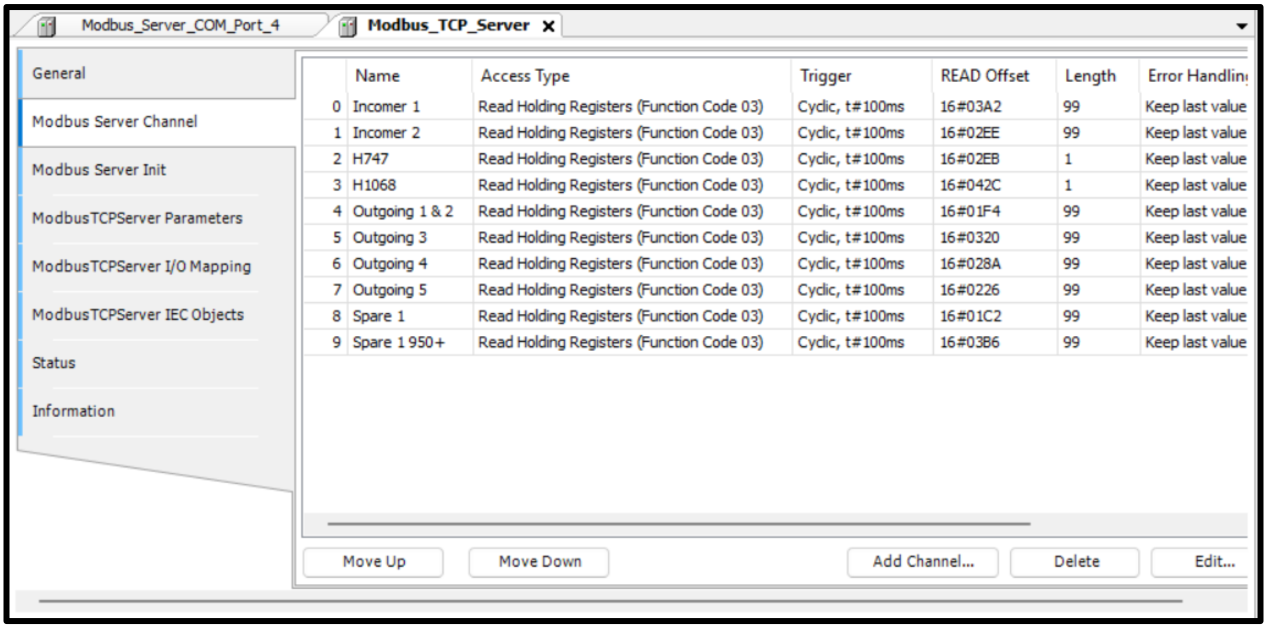
Figure 3: The provided image illustrates the Modbus RTU Serial configuration of the SCADA panel PLC, which acts as the dedicated gateway for high-reliability data acquisition from field devices. By utilizing Modbus RTU over an RS-485 serial interface, the system ensures the secure and stable transmission of cumulative energy values

3.2 Modbus TCP Communication

In addition to Modbus RTU, the SCADA panel PLC is configured to provide the same measurement data over Modbus TCP, enabling Ethernet-based communication for higher-level data access. While Modbus RTU is primarily used for cumulative energy data, Modbus TCP is utilized to transmit real-time electrical parameters that require faster update rates.

In the implemented system, the SCADA panel PLC functions as a Modbus TCP server, exposing internal registers that contain processed data obtained from multiple DPMs. These registers include parameters such as voltage, current, power, and power factor, which are continuously updated based on the latest readings from the field devices. The Wecon LX6C PLC operates as a Modbus TCP client, periodically reading these registers over the Ethernet network.

By using Modbus TCP for real-time electrical parameters, the system benefits from higher communication speed, reduced latency, and improved data refresh rates compared to serial communication. This approach complements the RTU-based energy data acquisition, resulting in a hybrid communication architecture where Modbus RTU ensures reliable acquisition of cumulative energy values, while Modbus TCP delivers fast and efficient access to instantaneous electrical parameters. The combined use of both protocols allows the Wecon PLC to receive a complete dataset without modifying the original multi-DPM RS-485 wiring, before forwarding the consolidated information to the HMI for visualization and MQTT-based transmission to the backend system.



	Name	Access Type	Trigger	READ Offset	Length	Error Handlin
0	Incomer 1	Read Holding Registers (Function Code 03)	Cyclic, t#100ms	16#03A2	99	Keep last value
1	Incomer 2	Read Holding Registers (Function Code 03)	Cyclic, t#100ms	16#02EE	99	Keep last value
2	H747	Read Holding Registers (Function Code 03)	Cyclic, t#100ms	16#02EB	1	Keep last value
3	H1068	Read Holding Registers (Function Code 03)	Cyclic, t#100ms	16#042C	1	Keep last value
4	Outgoing 1 & 2	Read Holding Registers (Function Code 03)	Cyclic, t#100ms	16#01F4	99	Keep last value
5	Outgoing 3	Read Holding Registers (Function Code 03)	Cyclic, t#100ms	16#0320	99	Keep last value
6	Outgoing 4	Read Holding Registers (Function Code 03)	Cyclic, t#100ms	16#028A	99	Keep last value
7	Outgoing 5	Read Holding Registers (Function Code 03)	Cyclic, t#100ms	16#0226	99	Keep last value
8	Spare 1	Read Holding Registers (Function Code 03)	Cyclic, t#100ms	16#01C2	99	Keep last value
9	Spare 1 950+	Read Holding Registers (Function Code 03)	Cyclic, t#100ms	16#03B6	99	Keep last value

Figure 4 Parameters used Modbus TCP Server

Figure 4: This image illustrates the Modbus TCP Server Channel configuration of the SCADA panel PLC, which serves as the operational bridge for high-speed, Ethernet-based data exchange. By utilizing Modbus TCP with a cyclic trigger of 100ms, the system achieves the rapid update rates necessary for real-time electrical parameters

3.3 MQTT Communications

For communication between the control layer and the backend server, the system employs MQTT (Message Queuing Telemetry Transport) due to its lightweight structure and suitability for IoT-based applications. The HMI, which is connected to the Wecon LX6C PLC, is equipped with built-in Wi-Fi and MQTT communication capability. After receiving processed energy data from the PLC, the HMI publishes the data to an MQTT broker over a wireless network.

The MQTT communication follows a publish–subscribe architecture, where the HMI acts as the publisher, and the backend system subscribes to predefined topics. Each topic represents specific energy parameters or monitoring locations, allowing structured and organized data transmission. This method ensures efficient use of network bandwidth and enables real-time data updates with minimal latency.

On the server side, Node-RED, running inside the virtual machine, functions as the MQTT subscriber. It listens to incoming MQTT messages, validates the data, and processes it before inserting the values into the MySQL database. The use of MQTT provides reliable, scalable, and near real-time communication between the field devices and the data management system, making it suitable for continuous energy monitoring applications.

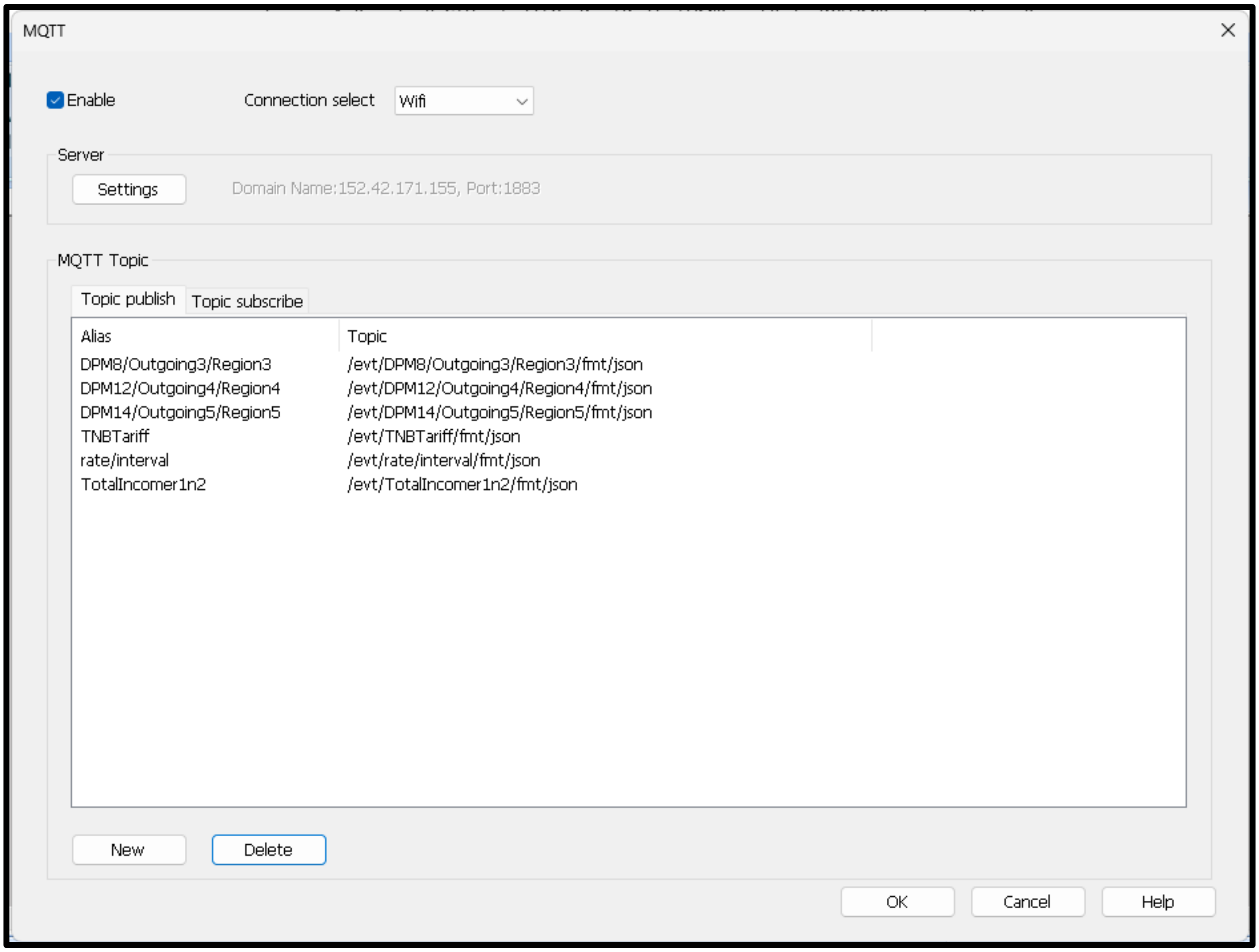


Figure 5 Multiple Topic Publish in MQTT

Figure 5 shows the MQTT configuration used to transmit power monitoring data over a Wi-Fi network. Multiple MQTT topics are defined to publish real-time electrical parameters, tariff information, and total energy consumption in JSON format. These topics enable structured data exchange between the PLC/HMI and the cloud server for further processing and visualization.

4. Backend System Architecture

All backend applications are hosted on a Tenaga Virtual Machine (VM) to ensure centralized management, scalability, and efficient resource utilization. Each application is containerized using Docker, providing isolation and ease of deployment. The container setup includes:

- **LAMP_app**: A container running a complete LAMP stack with Apache, PHP, MySQL, phpMyAdmin, and static HTML content.
- **Node-RED**: A dedicated container for receiving MQTT payloads, processing them, and inserting the data into the MySQL database.
- **Mosquitto**: A separate container acting as the MQTT broker for real-time message communication.

This architecture ensures that each application runs independently, avoids conflicts, simplifies updates, and improves overall system reliability.

4.1 Node-RED

Node-RED is used as the middleware platform in the Power Monitoring System to manage data communication between the HMI and the database. It runs inside the Tenaga virtual machine and acts as the central layer that receives, processes, and stores energy and electrical parameter data. The flow-based architecture of Node-RED allows reliable handling of continuous data streams while maintaining flexibility for system maintenance and future expansion.

In this system, Node-RED operates as an MQTT subscriber. The HMI publishes data via Wi-Fi after receiving and consolidating measurements from the Wecon LX6C PLC. These MQTT messages contain parameters collected from multiple digital power meters, including energy and other electrical values. Once the data is received, Node-RED performs basic validation to ensure that the payload structure and values are correct before further processing.

To control the data logging frequency and reduce database load, a fixed logging interval of **15 minutes** is implemented in the Node-RED flow. This ensures that data is stored at consistent intervals, which is sufficient for energy monitoring, trend analysis, and billing-related calculations. This interval-based approach prevents redundant data storage while maintaining meaningful historical records.

After passing the interval control, a function node is used to process the MQTT payload. The function node extracts the required parameters, assigns timestamps, and maps each value to the corresponding database field based on the predefined schema. The processed data is then sent to the MySQL node, which executes the insert operation and stores the data in the appropriate tables. Through this process, Node-RED ensures reliable and structured data transfer from the HMI to the database, supporting accurate visualization and analysis on the dashboard.

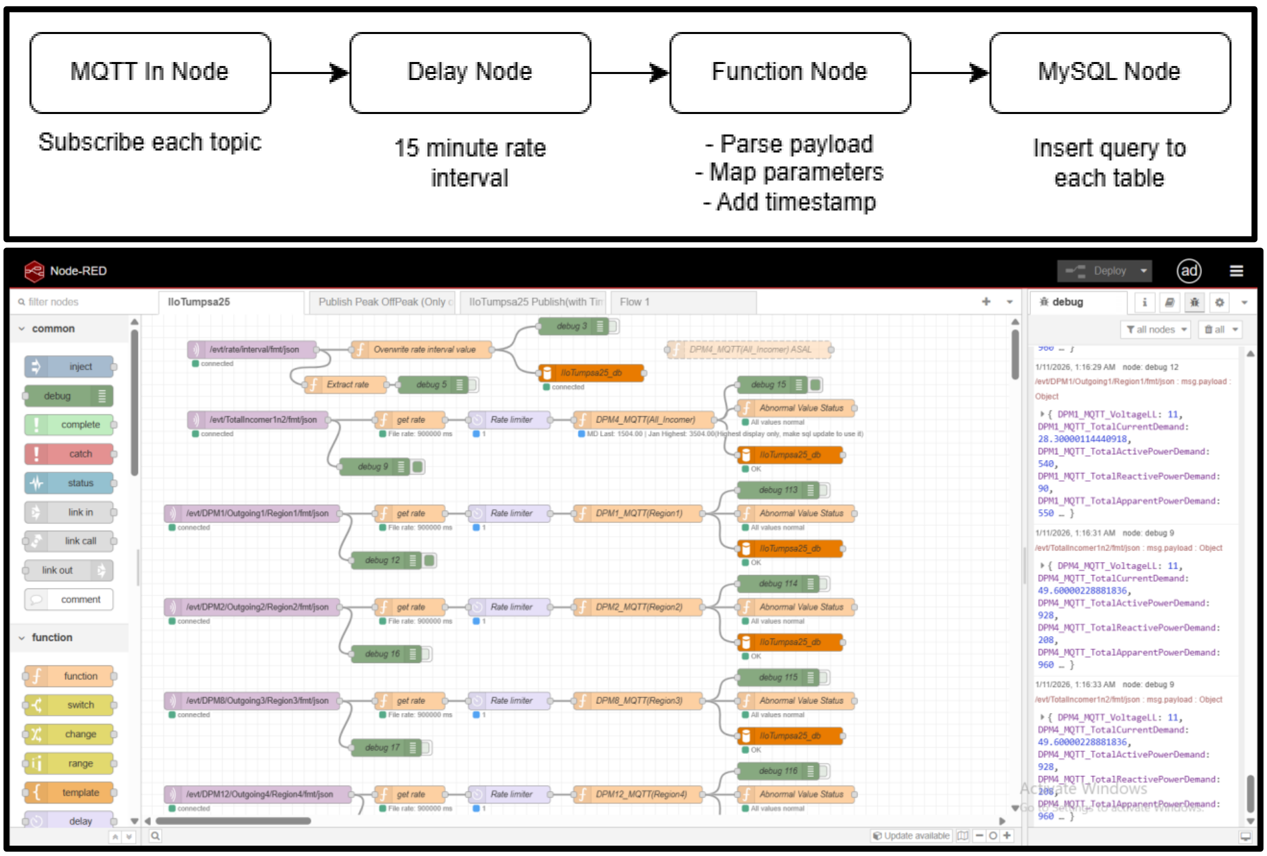


Figure 6 Flow in Node-RED

4.2 MySQL Database

The MySQL database is the central repository for the Power Monitoring System. It stores all energy and electrical parameters collected from the field devices, including both high-resolution raw data and daily aggregated data. Node-RED inserts the 15-minute data from the HMI into the raw tables, and scheduled events and triggers automatically summarize the raw data into the day tables. Billing calculations are performed based on the summarized data using the electricity tariffs stored in the tnbtariff table. This database structure ensures accurate energy monitoring, historical analysis, and cost estimation while maintaining performance and scalability.

4.2.1 Energy Table

- Incomer 1 Raw & Day
- Incomer 2 Raw & Day
- Incomer Total Day
- Region 1 Raw & Day
- Region 2 Raw & Day
- Region 3 Raw & Day
- Region 4 Raw & Day
- Region 5 Raw & Day

4.2.2 Parameter Table

- `userlogin` – stores user credentials and access levels
- `timeparameter` – stores scheduling and time-related configurations
- `tnbtariff` – stores electricity tariff rates for billing calculations

4.2.3 Table Structure

1) Raw Table for Incomer & Region

The **raw table** stores detailed electrical measurements collected at specific points in time. Each record is identified by the **Timestamp**, which is set as the **primary key** to ensure there are no duplicate entries. The table also includes the **Date**, which is automatically generated for easier daily grouping. All electrical data, including energy, voltage, current, power, and power factor, are stored as **decimal (18,2)**. Using the decimal type ensures the values are **accurate, readable, and safe for calculations**, avoiding rounding errors or incorrect results that can occur with float or other data types. This structure allows the raw table to serve as a reliable source for analysis, reporting, and dashboard visualization of energy consumption and power quality.

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra
1	Timestamp	datetime			No	*None*		
2	Date	date			No	curdate()		DEFAULT_GENERATED
3	TotalImportActiveEnergy	decimal(18,2)			No	0.00		
4	TotalImportActiveEnergyPeakHour	decimal(18,2)			No	0.00		
5	TotalImportActiveEnergyOffPeakHour	decimal(18,2)			No	0.00		
6	DiffTotalImportActiveEnergyPeakHour	decimal(18,2)			No	0.00		
7	DiffTotalImportActiveEnergyOffPeakHour	decimal(18,2)			No	0.00		
8	VoltageLL	decimal(18,2)			No	0.00		
9	CurrentLN1	decimal(18,2)			No	0.00		
10	CurrentLN2	decimal(18,2)			No	0.00		
11	CurrentLN3	decimal(18,2)			No	0.00		
12	TotalCurrentDemand	decimal(18,2)			No	0.00		
13	TotalActivePowerDemand	decimal(18,2)			No	0.00		
14	TotalReactivePowerDemand	decimal(18,2)			No	0.00		
15	TotalApparentPowerDemand	decimal(18,2)			No	0.00		
16	Frequency	decimal(18,2)			No	0.00		
17	TotalPowerFactor	decimal(18,2)			No	0.00		
18	TotalActivePower	decimal(18,2)			No	0.00		
19	TotalReactivePower	decimal(18,2)			No	0.00		
20	TotalApparentPower	decimal(18,2)			No	0.00		

Figure 7 Table Structure for Incomer & Region Raw table

2) Day Table for Incomer & Region

The **day table** stores **daily aggregated electrical data** derived from the raw table. Each record is identified by a **Timestamp** as the primary key, with the **Date** also indexed for easy grouping and querying. The electrical measurements, such as voltage, current, power, and energy, are **cleaned and summarized** from the raw table to represent daily values, ensuring consistent and accurate data. In addition, this table includes **calculated fields** such as energy costs (`CostPeakEnergy`, `CostOffPeakEnergy`, `CostEnergyTotal`) and environmental metrics (`SolarEnergy`, `CO2Emission`, `SolarCost`) based on the daily energy consumption. All electrical values are stored as **decimal** to maintain precision and prevent calculation errors, making the table reliable for reporting, analysis, and dashboard visualizations of daily energy usage and costs.

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra
1	**Timestamp**	datetime			No	*None*		
2	**Date**	date			No	curdate()		DEFAULT_GENERATED
3	**SumDiffTotalImportActiveEnergyPeakHour**	decimal(18,2)			No	0.00		
4	**SumDiffTotalImportActiveEnergyOffPeakHour**	decimal(18,2)			No	0.00		
5	**VoltageLL**	decimal(10,2)			No	0.00		
6	**CurrentLN1**	decimal(10,2)			No	0.00		
7	**CurrentLN2**	decimal(10,2)			No	0.00		
8	**CurrentLN3**	decimal(10,2)			No	0.00		
9	**TotalCurrentDemand**	decimal(10,2)			No	0.00		
10	**TotalActivePowerDemand**	decimal(18,2)			No	0.00		
11	**TotalReactivePowerDemand**	decimal(18,2)			No	0.00		
12	**TotalApparentPowerDemand**	decimal(18,2)			No	0.00		
13	**Frequency**	decimal(10,2)			No	0.00		
14	**TotalPowerFactor**	decimal(10,2)			No	0.00		
15	**TotalActivePower**	decimal(18,2)			No	0.00		
16	**TotalReactivePower**	decimal(18,2)			No	0.00		
17	**TotalApparentPower**	decimal(18,2)			No	0.00		
18	**SolarEnergy**	decimal(18,2)			No	0.00		
19	**CO2Emission**	decimal(18,2)			No	0.00		
20	**CostPeakEnergy**	decimal(18,2)			No	0.00		
21	**CostOffPeakEnergy**	decimal(18,2)			No	0.00		
22	**CostEnergyTotal**	decimal(18,2)			No	0.00		
23	**SolarCost**	decimal(18,2)			No	0.00		

Figure 8 Table Structure for Incomer & Region Day table

3) Day Table for Incomer Total

The **Incomer Total Table** is designed to store **aggregated daily electrical data** total from Incomer 1 and Incomer 2. Its structure is similar to the **day table**, containing cleaned and summarized measurements such as voltage, current, power, energy, and environmental metrics. All values are stored as **decimal** to ensure precision in calculations and avoid errors from floating-point data. Unlike the day table, this table includes **additional fields for billing calculations**, such as capacity, network, utility, renewable energy, and total costs, making it suitable for **TNB billing purposes**. The **Timestamp** is set as the primary key, and **Date** is indexed to prevent duplicate records and allow easy daily aggregation. This table serves as a reliable source for **financial reporting, energy cost analysis, and official utility billing**.

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra
1	**Timestamp**	datetime			No	*None*		
2	**Date**	date			No	curdate()		DEFAULT_GENERATED
3	**SumDiffTotalImportActiveEnergyPeakHour**	decimal(18,2)			No	0.00		
4	**SumDiffTotalImportActiveEnergyOffPeakHour**	decimal(18,2)			No	0.00		
5	**VoltageLL**	decimal(10,2)			No	0.00		
6	**CurrentLN1**	decimal(10,2)			No	0.00		
7	**CurrentLN2**	decimal(10,2)			No	0.00		
8	**CurrentLN3**	decimal(10,2)			No	0.00		
9	**TotalCurrentDemand**	decimal(10,2)			No	0.00		
10	**TotalActivePowerDemand**	decimal(18,2)			No	0.00		
11	**TotalReactivePowerDemand**	decimal(18,2)			No	0.00		
12	**TotalApparentPowerDemand**	decimal(18,2)			No	0.00		
13	**Frequency**	decimal(10,2)			No	0.00		
14	**TotalPowerFactor**	decimal(10,2)			No	0.00		
15	**TotalActivePower**	decimal(18,2)			No	0.00		
16	**TotalReactivePower**	decimal(18,2)			No	0.00		
17	**TotalApparentPower**	decimal(18,2)			No	0.00		
18	**SolarEnergy**	decimal(18,2)			No	0.00		
19	**CO2Emission**	decimal(18,2)			No	0.00		
20	**CostPeakEnergy**	decimal(18,2)			No	0.00		
21	**CostOffPeakEnergy**	decimal(18,2)			No	0.00		
22	**CostEnergyTotal**	decimal(18,2)			No	0.00		
23	**CostCapacity**	decimal(18,2)			No	0.00		
24	**CostNetwork**	decimal(18,2)			No	0.00		
25	**RebateTotal**	decimal(18,2)			No	0.00		
26	**CostTotal**	decimal(18,2)			No	0.00		
27	**CostRenewableEnergy**	decimal(18,2)			No	0.00		
28	**CostUtility**	decimal(18,2)			No	0.00		
29	**SolarCost**	decimal(18,2)			No	0.00		
30	**EstimatedCost**	decimal(18,2)			No	0.00		

Figure 9 Table Structure of Incomer Total Day table

4) Time Parameter

The **TimeParameter table** is used to store key configuration values for energy data processing. It contains parameters such as **PeakStart** and **PeakEnd**, which define the peak energy period, allowing the system to correctly assign raw energy data to either peak or off-peak columns. The table also includes a **rate** parameter, which sets a threshold for acceptable data loss. If the raw data exceeds this threshold, a **60/40 calculation** is applied to ensure the aggregated energy data remains accurate and reliable. This table provides a flexible way to manage energy calculation rules without changing the main raw or day tables.

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra
1	**id**	int			No	*None*		AUTO_INCREMENT
2	**ParamName**	varchar(50)	utf8mb4_0900_ai_ci		No	*None*		
3	**ParamValue**	varchar(50)	utf8mb4_0900_ai_ci		No	*None*		

Figure 10 Time Parameter in Database

5) TNB Tariff

The **TNB Tariff table** stores the official electricity rates and charges from TNB, which are used for billing calculations. It includes rates for **peak and off-peak energy, capacity, network, retailing, renewable energy, rebates, and solar energy**, as well as factors like **CO2 emission** and **solar power generation ratio**. This table allows the system to calculate close accurate electricity costs and billing amounts based on daily or total energy usage on the current month.

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra
1	**ID**	int			No	*None*		AUTO_INCREMENT
2	**ItemName**	varchar(100)	utf8mb4_0900_ai_ci		No	*None*		
3	**Value**	decimal(15,4)			No	*None*		
4	**Unit**	varchar(50)	utf8mb4_0900_ai_ci		Yes	*NULL*		

Figure 11 TNB Tariff in Database

6) User Login Information

The **UserLogin table** stores the login credentials for system users. It includes **UserID** and **UserPassword**, both stored as `varchar(12)`, which are required for authentication. This table ensures that only authorized users can access the system and its dashboards.

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra
1	**UserID**	varchar(12)	utf8mb4_0900_ai_ci		No	*None*		
2	**UserPassword**	varchar(12)	utf8mb4_0900_ai_ci		No	*None*		

Figure 12 User Information in Database

4.2.4 Trigger and Event Schedule

In our system database, we use a trigger and event schedule to make a calculation of the data inside the raw and day table to ensure data integrated on the html dashboard is clean and readable for visualization.

1) Trigger

   Here is the list of triggers we used on our database, each incomer and region table has the same trigger to ensure manageable and easy to troubleshoot

   - Peak and Off-peak hour window

     This trigger is used for allocating the energy data on the peak hour window or off-peak hour window. It will decide which column to be placed because the TNB bill splits the energy data into peak and off-peak hour cumulative data.

   - Calculate Energy Differentiate

     After the energy data allocated to its column, it will calculate the differentiate of the current data with previous data to get the actual energy usage at the time.

   - Data Lost Control

     This trigger allows the system to assume the data receive on the database into 60% for off-peak energy and 40% for peak energy. This to ensure the energy data is still reliable if the data is lost for more than 15 minutes or according to the rate interval set on the Node-Red.

   - Sum Daily Energy Data

     From the calculated data of energy on the raw table, it will sum up the energy data daily and store it in the day table. It also finds the maximum demand of the active power; the other electrical parameter will be inserted into the day table, same as the raw table.

- Calculate Cost

  After daily summation data, it will calculate the billing cost according to the TNB billing and get the daily estimation cost until the last day of the month and will reset for next month
- Incomer Total Summation

  This trigger is used to sum and average the required data on the incomer 1 and incomer 2 to get incomer total.

2) Event Schedule

   In our database, we also used events scheduled to ensure the data that are not reliable or missing can be resolved to ensure the data is reliable and accurate. Here is the list of events that we use. All events configured as recurring events and executed every 5 seconds to check error all the time.

- Recalculate Energy Cost

  This event is used for recalculating the required billing cost if the maximum demand is changed or have any other data missing on the table.

- Update Maximum Demand

  This event used for always updating the maximum demand of the active power to ensure the calculation of the billing cost is always close to accurate and still reliable until the last day of the month.

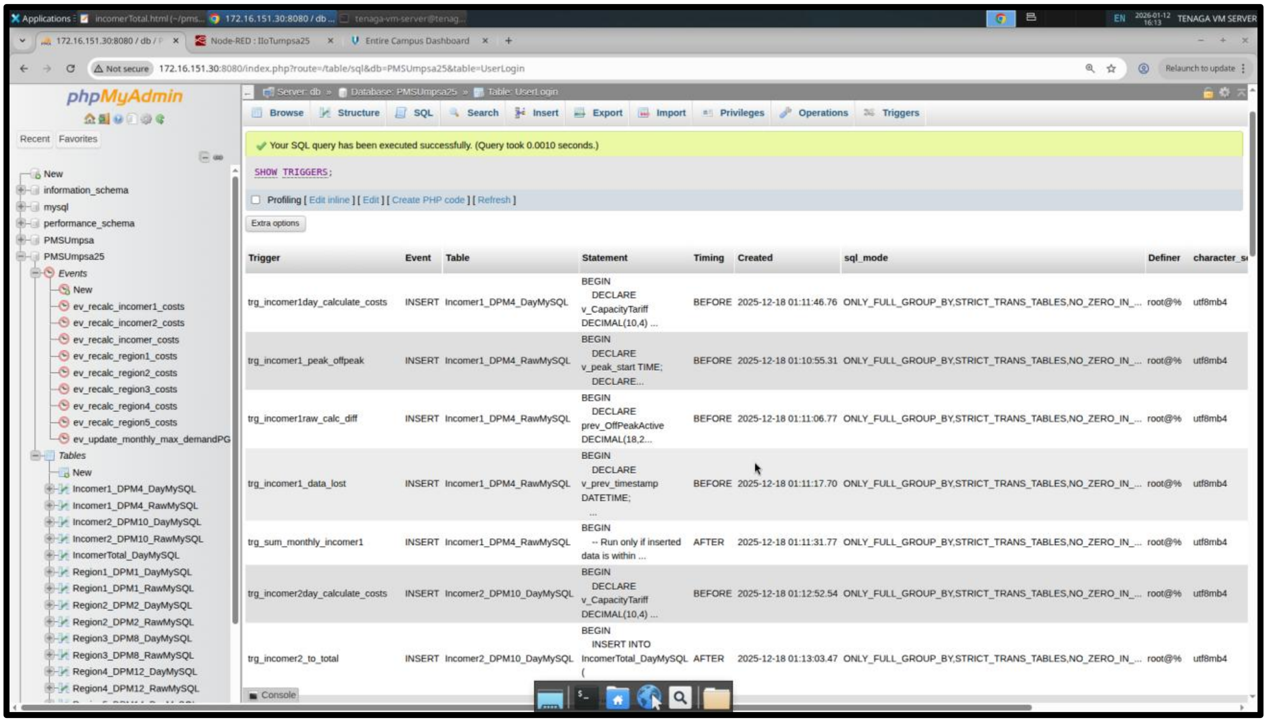


Figure 13 Trigger and Event on MySQL Database

5. API Integration

5.1 Overview

The API in the Power Monitoring System acts as a bridge between the **MySQL database** and the **HTML dashboard**, allowing secure and structured access to energy and electrical data. The API is implemented using **PHP (`api.php`)** and is hosted inside the same virtual machine as Node-RED and MySQL. It enables the dashboard to retrieve both real-time and historical data for visualization, reporting, and billing analysis.

5.2 Role and Functionality

The API serves several purposes in the system:

- Fetching **raw and day table data** from the MySQL database.
- Performing **filtering and aggregation** of data as needed by the dashboard.
- Providing data in a **structured format (JSON)** for frontend consumption.
- Supporting **real-time and historical data requests** for energy parameters, maximum demand, power factor, and billing information.
- Ensuring **secure access** to the database without exposing it directly to the web interface.

5.3 API Request and Response Flow

- The **HTML dashboard** sends a request to `api.php` specifying the required parameters (date range, incomer, or region).
- The **API receives the request**, validates it, and executes SQL queries on the MySQL database.
- The **API formats the results** into JSON, including the requested energy parameters, calculated totals, or billing cost.
- The **dashboard receives the JSON data** and uses it to populate line charts, tables, or other visualization widgets.

5.4 Key Features

- Supports requests for **specific incomers or regions**, enabling selective data retrieval.
- Provides **daily, monthly, or custom-period summaries** from the day tables.
- Returns **billing cost calculations** based on the `tnbtariff` table and daily energy consumption.
- Optimized to handle **large datasets** efficiently without overloading the server.
- Enables **scalability**, allowing future integration with other systems or mobile applications.

5.5 Security Considerations

- The API uses a **separate `config.php` file** to store MySQL credentials, ensuring that sensitive information such as database username, password, and host is not hardcoded directly in the main API (`api.php`). This improves maintainability and reduces security risks.
- The API includes **input validation and sanitization** to prevent SQL injection and unauthorized access.
- Only authenticated users can access certain endpoints, with user credentials managed through the `userlogin` table.
- Data is transmitted in a structured format (JSON), allowing secure and standardized integration with the HTML dashboard while keeping database details hidden.

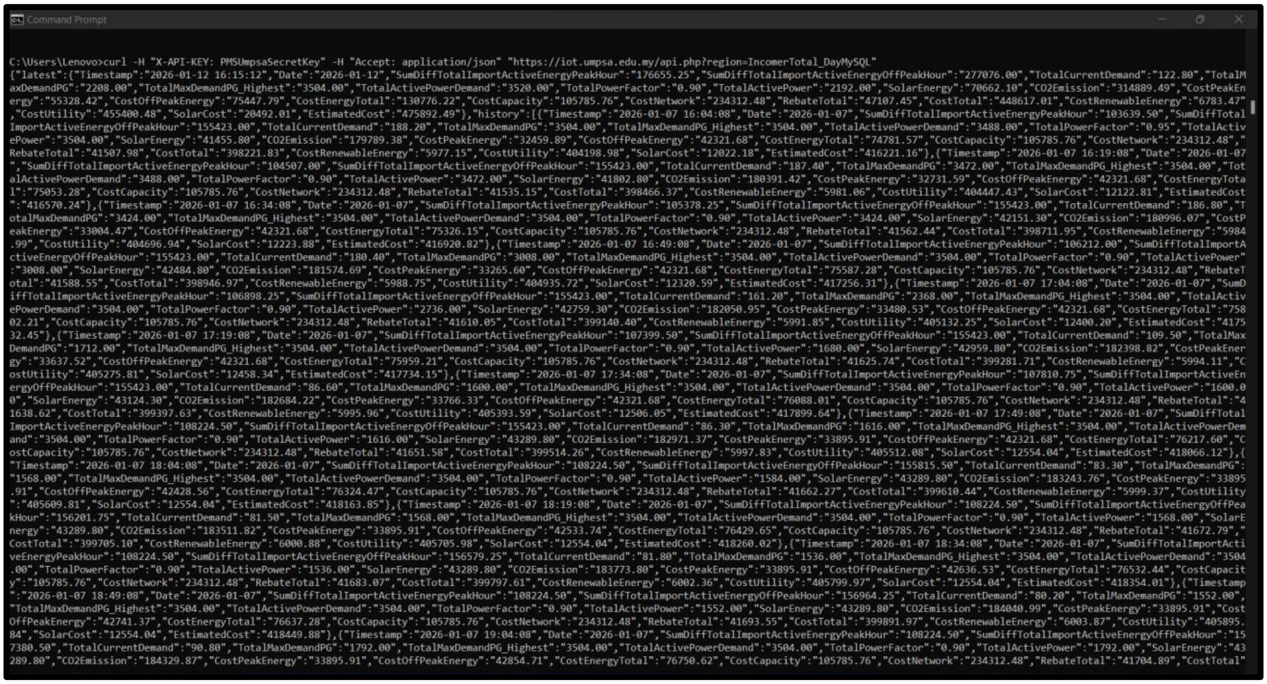


Figure 14 Output of the Data formatted JSON

6. Dashboard & Data Visualization

For the dashboard and data visualization, we used HTML, CSS, and JavaScript to develop the required dashboard page of the system to make an understandable and interactive dashboard for the user that has a background of non-technical. For every table of the raw and day table, it has its own page to visualize the required data to the user.

6.1 Dashboard Features

- **Real-time monitoring:** Displays instantaneous voltage, current, power, and power factor.
- **Energy consumption trends:** Line charts showing active and reactive energy over time.
- **Maximum demand visualization:** Highlights peak power usage for each incomer or region.
- **Energy consumption comparison:** Showing the proportion of total energy usage for each region, based on aggregated data from day tables.
- **Billing and cost analysis:** Calculates estimated electricity cost using data from the day table and `tnbtariff`.
- **User management:** Supports login and access control using the `userlogin` table.

6.2 Data Flow

- The **dashboard sends a request** to `api.php` specifying the required incomer/region, date range, and data type (raw or aggregated).
- The **API fetches data** from the MySQL database using the credentials stored in `config.php`.
- The **API formats the data** into JSON and returns it to the dashboard.
- The dashboard **renders charts, tables, and indicators** using JavaScript and charting libraries for visualization.

6.3 Visualization Component

- **Line Charts:** For historical maximum demand trend and power factor every 15 minutes.
- **Bar Charts:** Displaying the historical daily energy consumption and estimated cost.
- **Numeric Indicators:** For real-time energy usage, maximum demand, power consumption, and current estimated cost and forecasted this month's cost.
- **Tables:** Display the detailed breakdown of the estimated cost billing according to TNB billing
- **Pie Chart:** For visualize the comparison of the energy consumption by each region in real-time

6.4 Dashboard Page Functionality

Here is the list of pages on the dashboard

6.4.1 Login

The login page is used to allow users to access the system securely. Users must enter their **username and password** to log in. When the login button is clicked, the system checks the entered information with the data stored in the database. If the username and password are correct, the user is allowed to enter the system and is redirected to the dashboard. If the login details are incorrect, an error message will be displayed to inform the user. This helps prevent unauthorized access. The login page ensures that only authorized users can access the system and protect system data.

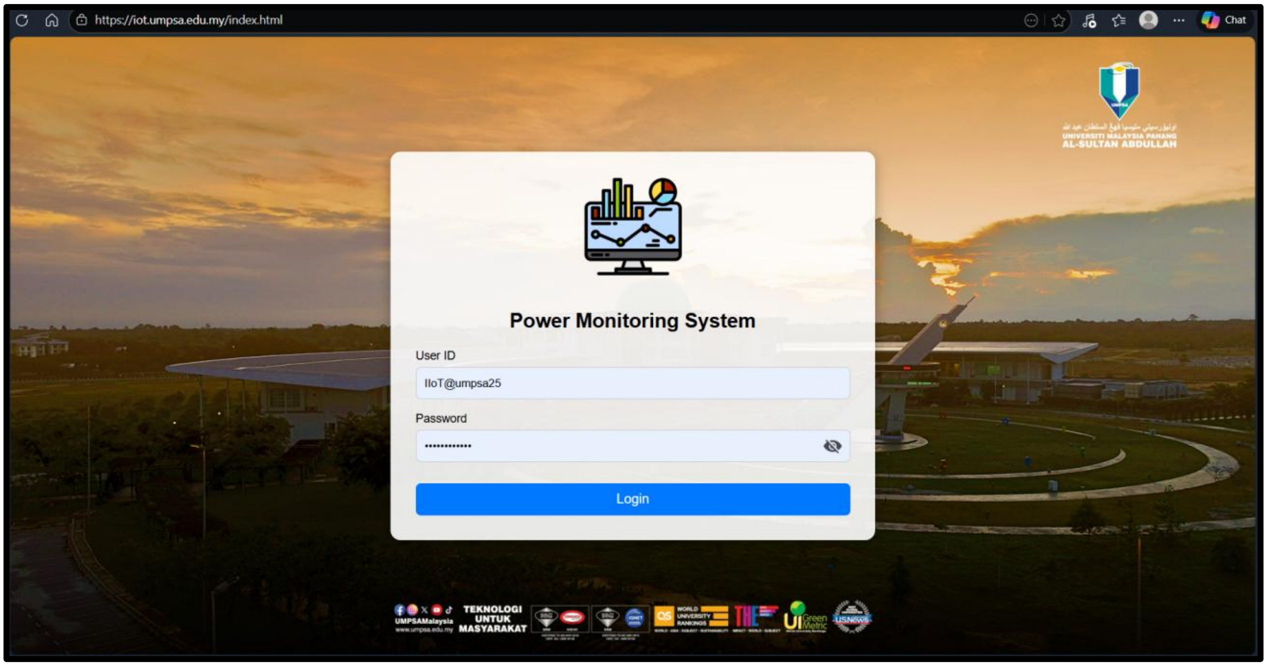


Figure 15 Login Page

6.4.2Home

The Home Page is the main dashboard of the Power Monitoring System. It provides an overall view of energy usage for the entire campus and individual regions in real time. The page displays an interactive map showing the campus layout with markers that represent different buildings and regions. Users can easily identify power usage locations and view regional boundaries on the map. On the left side, the **Entire Campus Summary** panel shows total energy usage, peak hour and off-peak hour consumption, carbon emission, and estimated electricity cost. This helps users understand the overall energy performance of the campus. On the right side, the **Region Summary** panel displays detailed information for a selected region, including energy usage, estimated cost, energy comparison, and maximum demand. Users can select different regions on the map to view specific data. Charts are used to compare peak and off-peak energy usage, making the data easy to understand visually. The Home Page also includes system controls such as logout and color mode options. Overall, the Home Page allows users to monitor energy consumption efficiently, compare regions, and make informed decisions based on real-time data.

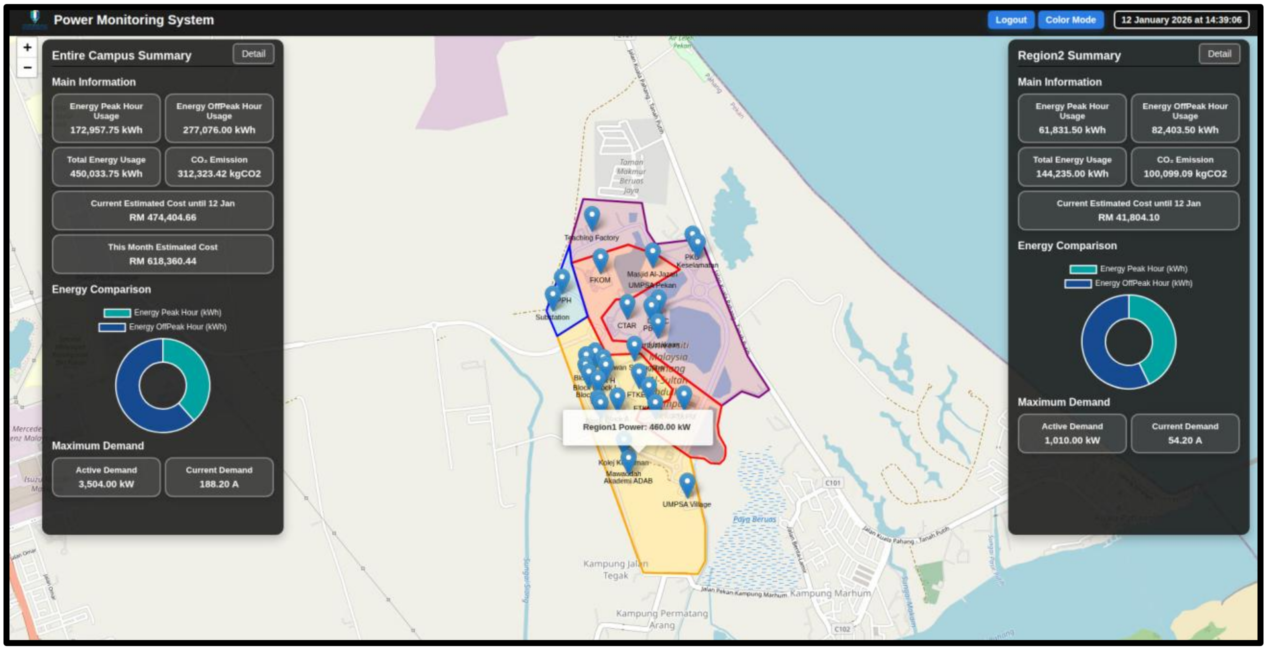
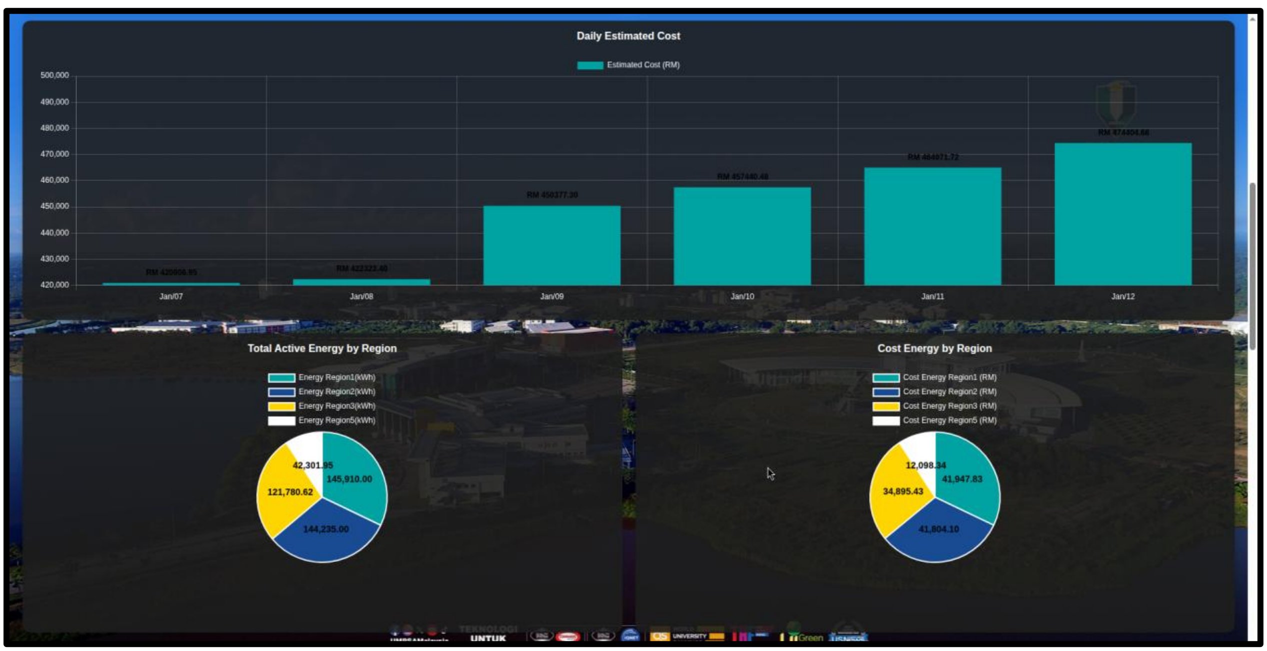


Figure 16 Home Page

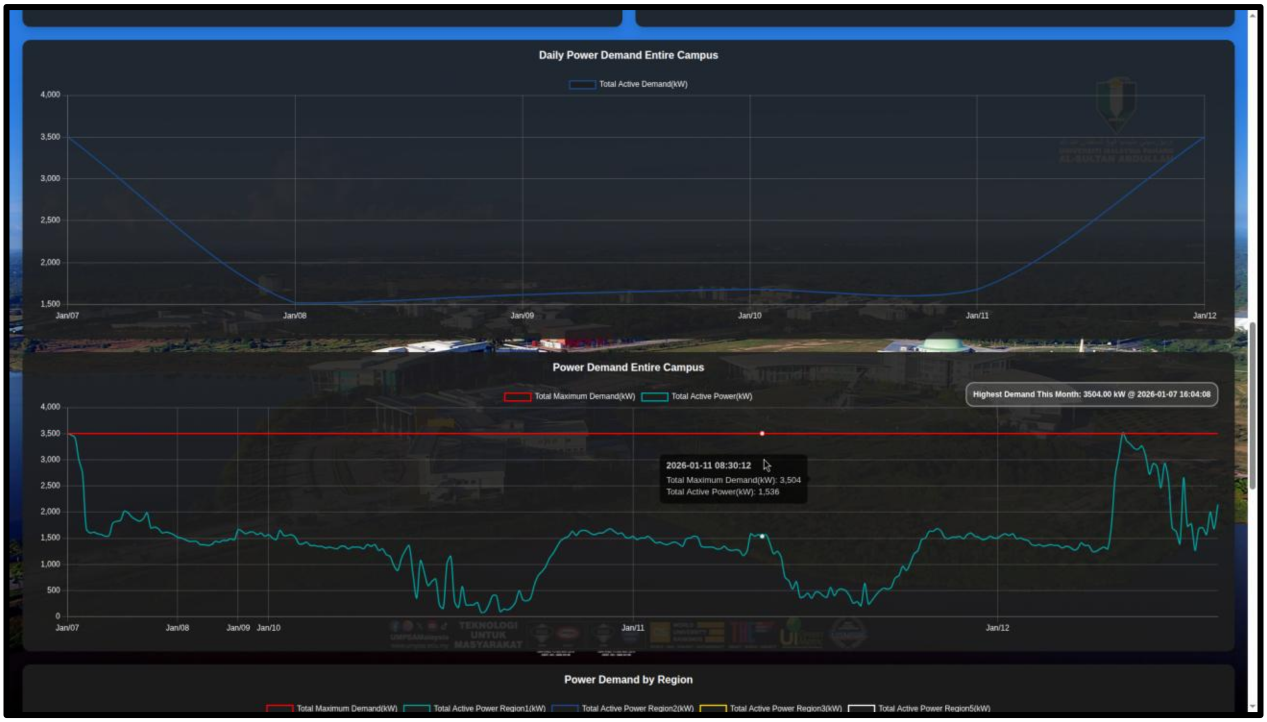
6.4.3 Entire Campus Dashboard

The **Entire Campus Detail Page** provides a comprehensive view of energy consumption, cost, and power performance for the whole campus in a single dashboard. This page displays an energy bar chart to show peak and off-peak energy usage over time, along with a pie chart that compares the proportion of peak and off-peak energy consumption for the entire campus. An estimated cost breakdown table is included to explain how the total electricity cost is calculated. A daily estimated cost bar chart is used to monitor daily cost trends. The page also includes pie charts to compare energy usage and energy cost by region, helping users identify high-consumption areas. In addition, daily power demand is visualized using line charts to highlight maximum demand points, while a power factor line chart is provided to monitor power efficiency. Users can navigate the data to view historical records filtered by 7 days, and all displayed data can be exported to Excel and PDF formats for reporting and further analysis.

Document Title		Power Monitoring System UMPSA			
		Ilot Substation			
Doc Type	Word	Document type	1	Date	19 December 2025



Document Title		Power Monitoring System UMPSA			
		Ilot Substation			
Doc Type	Word	Document type	1	Date	19 December 2025



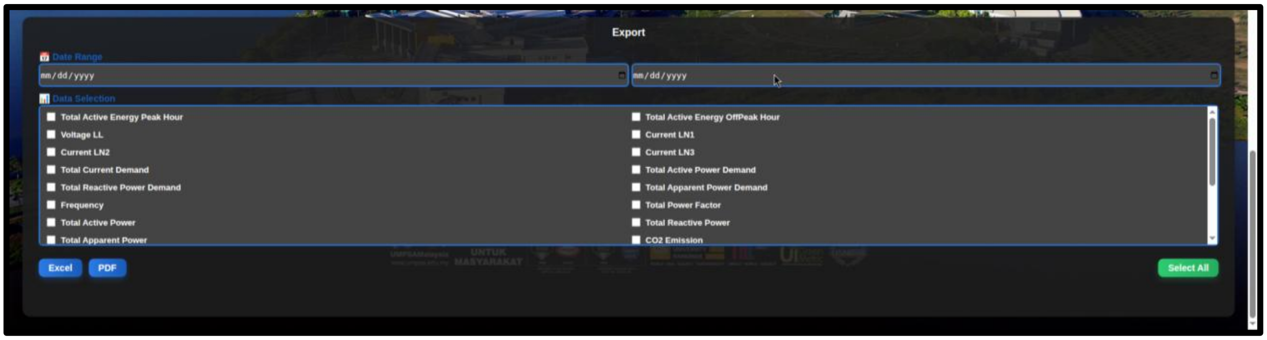


Figure 17 Entire Campus Dashboard

6.4.4 Incomer & Region Dashboard

The **Incomer and Region Dashboard** consist of seven pages, which include **Incomer 1**, **Incomer 2**, and **Region 1 to Region 5**. All seven pages share the same dashboard design to ensure consistency and ease of use. Each page displays a daily energy bar chart to show energy consumption trends, a pie chart to compare peak and off-peak energy costs, a power demand line chart to monitor demand patterns, and a power factor line chart to evaluate power quality and efficiency. An export function is provided to allow users to download data for reporting purposes. In addition, a data navigation function is available to view historical data. These dashboards are used to view detailed energy, cost, and power information for each individual incomer and region.

Document Title		Power Monitoring System UMPSA			
		Ilot Substation			
Doc Type	Word	Document type	1	Date	19 December 2025

Figure 18 Incomer & Region Dashboard

7. Conclusion

The development of the IIoT-based Power Monitoring System for UMPSA represents a significant leap from reactive utility management to a proactive, data-driven operational model. By successfully integrating industrial hardware with modern cloud technologies, the project establishes a seamless flow of information from the campus high-voltage incomers directly to the management desktop.

The Integrated Ecosystem

The strength of this system lies in its multi-layered integration:

- **Hardware Integration**: The system bridges the gap between field-level electrical infrastructure and digital control. By utilizing Digital Power Meters (DPMs) connected via Modbus RTU and TCP, the system ensures that critical energy data is captured with high precision and reliability.
- **Protocol Synergy**: The hybrid use of Modbus for local industrial communication and MQTT for cloud transmission demonstrates a robust architecture. This allows the WECON PLC and HMI to act as intelligent gateways, translating complex industrial signals into lightweight, web-friendly data packets.
- **Data Intelligence**: Through the orchestration of Node-RED and MySQL, the system transforms raw electrical pulses into actionable intelligence. The backend doesn't just store data; it organizes it into "Raw" and "Day" structures, allowing for both real-time troubleshooting and long-term trend analysis.
- **Managerial Empowerment**: The PHP-based API and HTML Dashboard serve as the final link, converting technical parameters into financial insights. By incorporating TNB tariff structures, the system provides UMPSA management with the ability to forecast costs and monitor maximum demand, directly addressing the challenge of unpredicted utility expenditures.